

React JS

React JS is a JavaScript library used to build user interfaces for websites.

write react CDN on google if you want react code

<https://unpkg.com/react@18/umd/react.development.js> open this link you will see js in optimised way

React is javascript object

```
> React
< {Children: {...}, Fragment: Symbol(react.fragment), Profiler: Symbol(react.profiler), Component: f, PureComponent: f, ...}
> typeof React
< 'object'
> typeof ReactDOM
< 'object'
```

1. Why are there two scripts in React?

Why not use a single file?

Because React supports multiple platforms:

- react-dom → for websites
- react-native → for mobile apps
- react-three-fiber → for 3D apps

Same react core, different renderers

So React is separated by design.

They are not different Reacts

They are parts of the same React

- react.js → thinks (logic, components)
- react-dom.js → shows React on the web page

They work together.

React is one, but its work is split into two files: one for logic and one for showing it in the browser.

2. React 17 – Benefits

React 17 was made to upgrade safely.

- Easy to upgrade from old React
- No big changes → old code still works
- No need to write import React from 'react' in every file
- Useful for big apps

React 17 focus: stability, not new features

3.React 18 – Benefits

React 18 makes apps faster and smoother

1.Faster performance

- React does fewer re-renders
- App feels smooth

2.Better user experience

- UI does not freeze during heavy work

3.Automatic batching

- Multiple state updates = only one render

4.New features

- useTransition → heavy work in background
- useDeferredValue → smooth search & filtering

5.Modern root API

- Uses createRoot() for better performance

React 17 vs React 18

React 17	React 18
Safe upgrade	Faster app
No new features	Many new features
Stable	Modern & powerful

4.What is CDN?

CDN (Content Delivery Network) is a network of servers placed in many locations around the world.

It delivers website content (images, CSS, JS, videos) from the nearest server to the user.

static data will store in CDN , but comments like store in main server bcz they are dynamic

Why do we use CDN?

We use CDN to make websites:

- Faster
- More reliable
- Able to handle more users

What problem does CDN solve?

Without CDN (Problem)

- All users get data from one main server
- Far users get slow loading
- Server can be overloaded
- Website may crash

With CDN (Solution)

- Content comes from nearest server
- Faster loading time
- Less load on main server
- Website works even during high traffic

What happens if CDN is not present?

- Website loads slow
- Users far from server face delay
- Main server gets heavy load
- Poor user experience
- Higher chance of downtime

Simple Real-Life Example

Think of Amazon delivery

- Without CDN → one warehouse for whole country
- With CDN → many warehouses in different cities

CDN is used to deliver website content faster by serving it from the nearest server and reducing load on the main server.

